

Guessing an Unfair Coin

Francesco De Comit 

The question

In front of you are three coins, identical in appearance. However, while two of them are fair coins, the third comes up Heads 75% of time. Your task is to identify which coin is the unfair one, but you may only perform a total of three coins flips ! (For each flip, you can flip any coin you want, taking any past flip results into account.) After those three flips, you must declare which coin you believe to be the unfair one.

Assuming you “play optimally”, what is the likelihood that you will correctly identify the unfair coin after your three flips ?

In order to generalize, I will assume that the probability of coming up Heads is $p > 0.5$.

My solution

For the three flips I am allowed to perform, I choose to flip each coin once. Naming the unfair coin A , you have three equiprobable configurations :

- A, B, B named C_1 in the following.
- B, A, B named C_2
- B, B, A named C_3

Each configuration has eight possible outputs, with different probabilities according to the configuration.

Flips	C_1	C_2	C_3	$\frac{(C_1+C_2+C_3)}{3}$
Head,Head,Head	$p \times \frac{1}{2} \times \frac{1}{2}$	$p \times \frac{1}{2} \times \frac{1}{2}$	$p \times \frac{1}{2} \times \frac{1}{2}$	$\frac{p}{4}$
Head,Head,Tail	$p \times \frac{1}{2} \times \frac{1}{2}$	$p \times \frac{1}{2} \times \frac{1}{2}$	$(1-p) \times \frac{1}{2} \times \frac{1}{2}$	$\frac{p+1}{4 \times 3}$
Head,Tail,Head	$p \times \frac{1}{2} \times \frac{1}{2}$	$(1-p) \times \frac{1}{2} \times \frac{1}{2}$	$p \times \frac{1}{2} \times \frac{1}{2}$	$\frac{p+1}{4 \times 3}$
Head,Tail,Tail	$p \times \frac{1}{2} \times \frac{1}{2}$	$(1-p) \times \frac{1}{2} \times \frac{1}{2}$	$(1-p) \times \frac{1}{2} \times \frac{1}{2}$	$\frac{2-p}{4 \times 3}$
Tail,Head,Head	$(1-p) \times \frac{1}{2} \times \frac{1}{2}$	$p \times \frac{1}{2} \times \frac{1}{2}$	$p \times \frac{1}{2} \times \frac{1}{2}$	$\frac{p+1}{4 \times 3}$
Tail,Head,Tail	$(1-p) \times \frac{1}{2} \times \frac{1}{2}$	$p \times \frac{1}{2} \times \frac{1}{2}$	$(1-p) \times \frac{1}{2} \times \frac{1}{2}$	$\frac{2-p}{4 \times 3}$
Tail,Tail,Head	$(1-p) \times \frac{1}{2} \times \frac{1}{2}$	$(1-p) \times \frac{1}{2} \times \frac{1}{2}$	$p \times \frac{1}{2} \times \frac{1}{2}$	$\frac{2-p}{4 \times 3}$
Tail,Tail,Tail	$(1-p) \times \frac{1}{2} \times \frac{1}{2}$	$(1-p) \times \frac{1}{2} \times \frac{1}{2}$	$(1-p) \times \frac{1}{2} \times \frac{1}{2}$	$\frac{1-p}{4}$
Total proba	1	1	1	1

If the flip result is either Head,Head,Head of Tail,Tail,Tail, the best guess is random, and you make the correct guess with probability $\frac{1}{3}$. This case occurs with probability $\frac{1}{4}$.

Two Heads, One Tail

If the result is Head,Head,Tail :

- If we are in configuration C_1 , the probability of obtaining this draw is $\frac{p}{4}$.
- Same if we are in configuration C_2 .
- In configuration C_3 , the probability of this draw is $\frac{(1-p)}{4}$

So, if we observe Head,Head,Tail there is a chance equal to :

$$\frac{1}{3} \times \frac{p}{4} \times \frac{4 \times 3}{p+1} = \frac{p}{p+1}$$

to be in configuration C_1 .

The same chance if we are in C_2 , and a probability of $\frac{1-p}{p+1}$ of being in C_3 .

Consider the two configurations with Head : we are in one of those configurations with probability $\frac{2 \times p}{p+1}$.

Choose one of the coin which flips to Head gives you a chance of $\frac{p}{p+1}$ of guessing right.

And $\frac{p}{p+1} \geq \frac{1}{3}$ as soon as $p \geq \frac{1}{2}$.

The same reasoning holds for all the draws with two heads

Two Tails, One Head

Let's look at draw with two tails, a case which occurs with probability $\frac{2-p}{4}$.

Take for example Head,Tail,Tail

- If we are in configuration C_1 , the probability of obtaining this draw is $\frac{p}{4}$.
- In configuration C_2 , the probability of this draw is $\frac{(1-p)}{4}$
- In configuration C_3 , the probability of this draw is $\frac{(1-p)}{4}$

Let us compute the probability of being in each configurations if we observe Head,Tail,Tail.

- For configuration C_1 : $\frac{p}{4} \times \frac{4}{2-p} = \frac{p}{2-p}$.
- For configurations C_2 and C_3 : $\frac{1-p}{4} \times \frac{4}{2-p} = \frac{1-p}{2-p}$.

If you identify the first coin as being the unfair one, you will guess right with probability $\frac{p}{2-p}$, which is greater than $\frac{1}{3}$ as soon as $p \geq \frac{1}{2}$

Applying the same reasoning for all the draws with two tails gives the same result.

Conclusion

- In case of Head, Head,Head or Tails,Tails,Tails, pick a coin at random, with a probability of $\frac{1}{3}$ of winning.

- When the draw contains two heads, pick on of the head coin at random, leading to a probability of $\frac{p}{p+1}$ of winning.
- When the draw contains two tails, choose the head, with a probability of $\frac{p}{2-p}$ of winning.

In conclusion, this strategy is equivalent to random guessing in $\frac{1}{4}$ of the cases, and better than random in all other cases. More precisely, the probability can be computed the following way :

- Head, Head, Head: probability $\frac{p}{4}$, correct guess $\frac{1}{3}$
- Two heads, on tail : probability $\frac{p+1}{4}$, correct guess $\frac{p}{p+1}$
- One head, two tails : probability $\frac{2-p}{4}$, correct guess $\frac{p}{2-p}$
- Three tails : probability $\frac{1-p}{4}$, corect guess $\frac{1}{3}$

Which leads to a average probability of right guessing :

$$\begin{aligned} \frac{p}{4} \times \frac{1}{3} + \frac{p+1}{4} \times \frac{p}{p+1} + \frac{2-p}{4} \times \frac{p}{2-p} + \frac{1-p}{4} \times \frac{1}{3} \\ \frac{1}{4} \times \frac{1}{3} + 2 \times \frac{p}{4} \\ \frac{1}{12} + \frac{p}{2} \\ \frac{1+6 \times p}{12} \end{aligned}$$

Which is greater than $\frac{1}{3}$ when $p \geq 0.5$